ML Practical 5: Gradient Descent

Aim

To implement the Gradient Descent algorithm and find the minimum value of the function $y = (x + 3)^2$, starting from $x = 2$.

Theory

Gradient Descent is an optimization method used to find the minimum (lowest) value of a function. It starts with an initial value and makes small changes to that value step by step. The direction of each change is decided by the gradient, which tells us the slope of the function. The algorithm moves in the opposite direction of the gradient so that it can get closer to the minimum point.

Update Formula

$$\textbf{New } x = \textbf{Old } x - (\textbf{Learning Rate} \times \textbf{Gradient})$$

The **learning rate** controls the size of each step. A small learning rate takes smaller steps, while a large learning rate takes bigger steps. The gradient helps decide which direction to move.

Working of the Program

In this practical, the given function is $y = (x + 3)^2$. Its gradient is calculated as $2(x + 3)$. The program starts with $x = 2$, sets the learning rate to 0.1, and performs 50 iterations. During every iteration, it calculates the gradient and updates x using the formula. The values of x and the corresponding function values are stored to show the steps taken by the algorithm.

NumPy is used to generate points for drawing the function curve, and Matplotlib is used to display the curve, starting point, minimum point, and the Gradient Descent steps on a graph.

Result

After 50 iterations, the value of x approaches –3 and the function value approaches 0. Therefore, the minimum point is approximately (–3, 0).

Conclusion

Thus, the Gradient Descent algorithm is successfully implemented. It shows how repeated updates using the gradient help find the minimum value of the given function.